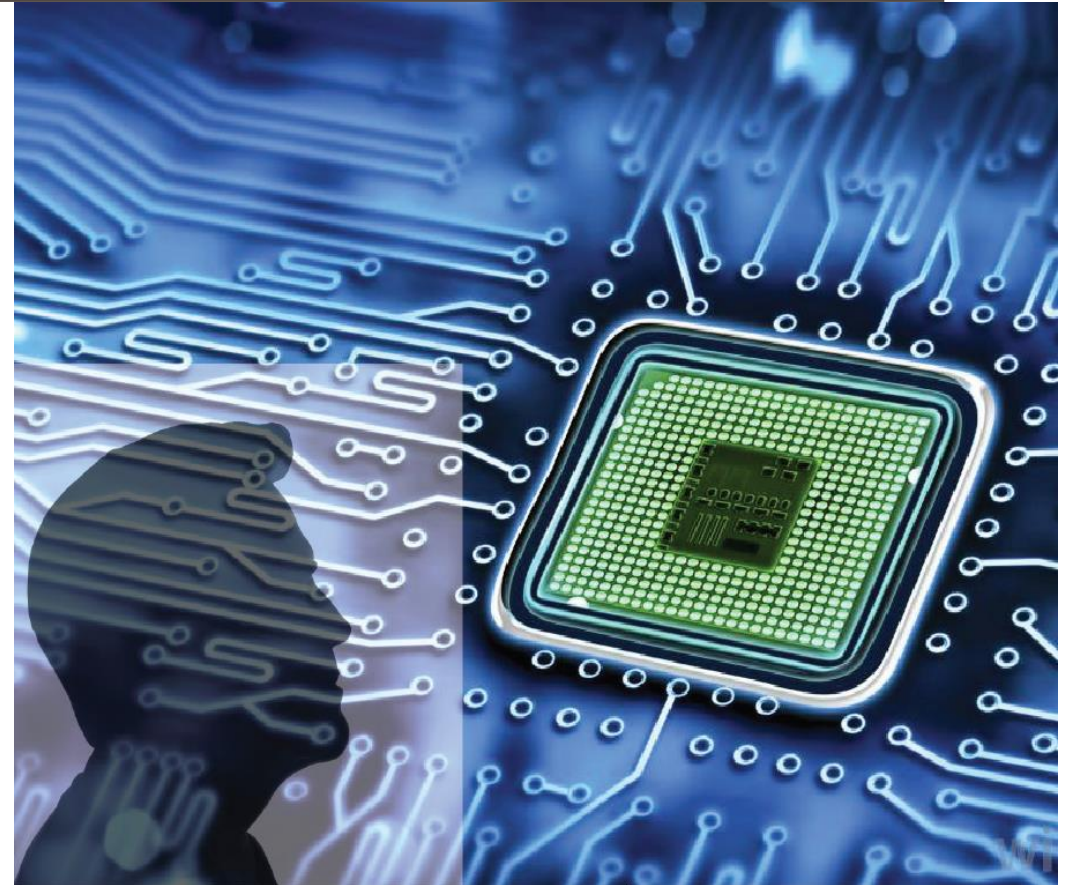


VLSI DESIGN FLOW

Week-1 Lecture-2 Introduction

Sneh Saurabh
14th August, 2024

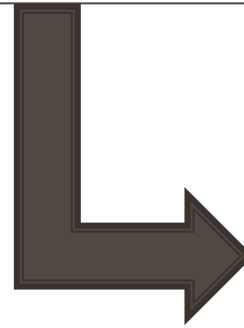


VDF: Revision/Announcements

- Project: Groups and Topics

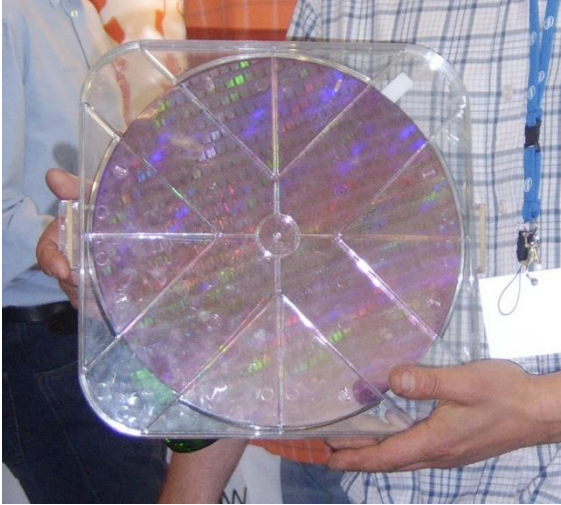


Introduction: Some Basic Concepts



Fabrication of Integrated
Circuits

Fabrication of Integrated Circuits: Silicon Wafer, Ingots



Source:
https://commons.wikimedia.org/wiki/File:CC_2008_Poland_Silicon_Wafer_1_edit.png
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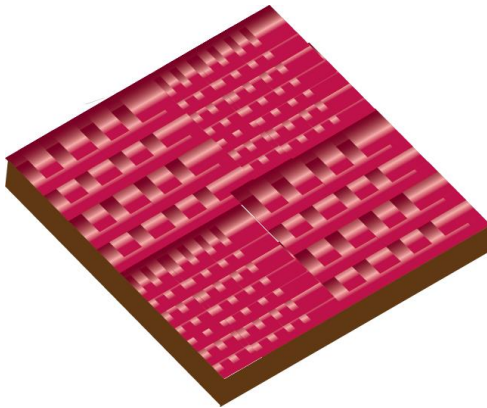
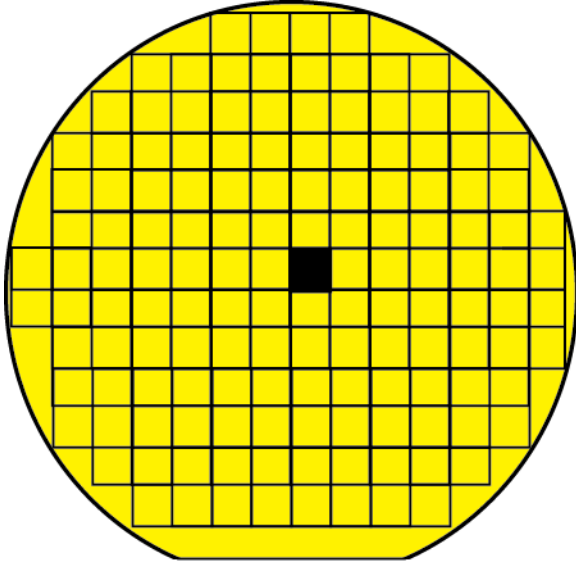
- A **silicon wafer** is a thin slice of silicon that serves as a substrate for the integrated circuit
- Integrated circuits are built over **silicon wafer**
- Currently, 300 *mm* wafer (*diameter* = 300 *mm*) are widely used

- A **silicon wafer** is sliced out from ***silicon ingots***
- Silicon ingots: massive cylindrical single crystal of silicon
- Silicon ingots are mostly prepared using Czochralski (CZ) process
 - Pure seed crystal is pulled out from a highly pure melted silicon at 1425°C



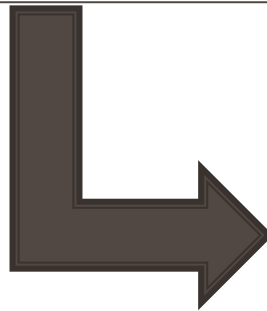
Source: https://commons.wikimedia.org/wiki/File:Monokristalines_Silizium_f%C3%BCr_die_Waferherstellung.jpg, German Wikipedia, original upload 7. Okt 2004 by [Stahlkocher](#)

Fabrication of Integrated Circuits: Dies and Chips



- Hundreds of rectangular shaped integrated circuits are fabricated on a single silicon wafer
- These slices of silicon wafer containing the complete circuit are called *dies*
- Dies are sliced out from silicon wafers after fabrication and testing
- After dies are sliced, they are encapsulated into a supporting case for protection against physical and chemical damage.
- Electrical contacts are made to connect externally.
- Cooling important
- Packaged dies are generally known as *chips*

VLSI Design Flow: A top-level perspective



Designing vs.
Fabrication

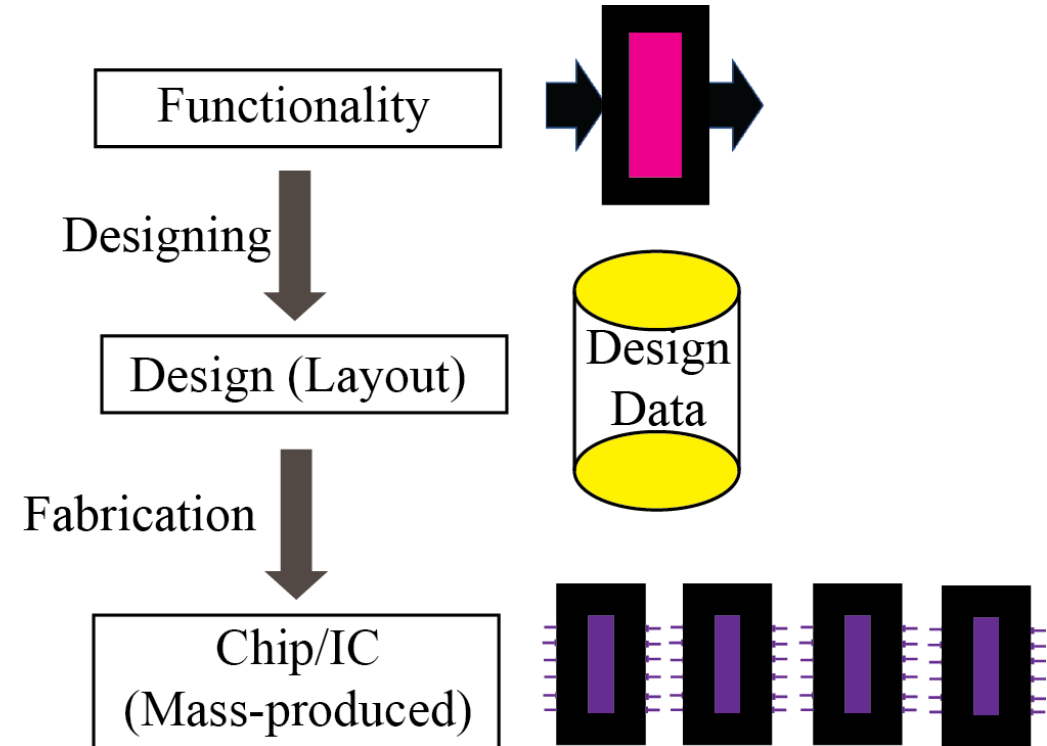
VLSI Design Flow: Semiconductor Foundries

Designing:

Finding a right set of *parameters* for the system, architecture, logic, gates, transistors, layout and devices to obtain the desired functionality of the chip applying suitable optimization.

Fabrication:

It involves **actual creation of integrated circuit** for a given design (layout of various layers) using steps like photolithography, ion-implantation etc.



VLSI Design Flow: Semiconductor Foundries

Foundry:

- Semiconductor manufacturing plant where the fabrication of integrated circuits is done.
- Cost of setting-up and maintaining is very high.
- Sustainable only when the **facilities of foundry are utilized close to their full potential**



GlobalFoundries Dresden Germany

Source:

https://commons.wikimedia.org/wiki/File:Aerial_photograph_of_Globalfoundries_Dresden.jpg Fensterblick., CC BY-SA 3.0

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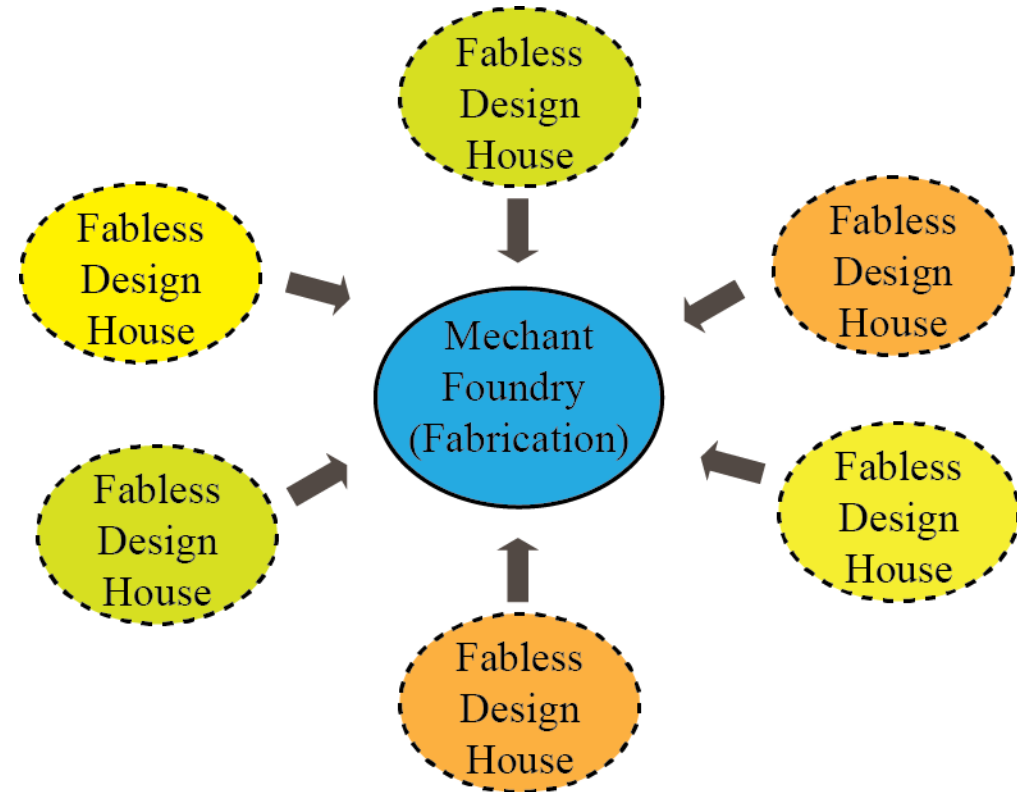
Semiconductor Industry: Business Model (1)

Fabless Design Companies:

- Only designing, fabrication is outsourced
- Do not require to setup and maintain costly foundries.
- Example: Qualcomm, Nvidia, etc.

Merchant Foundries:

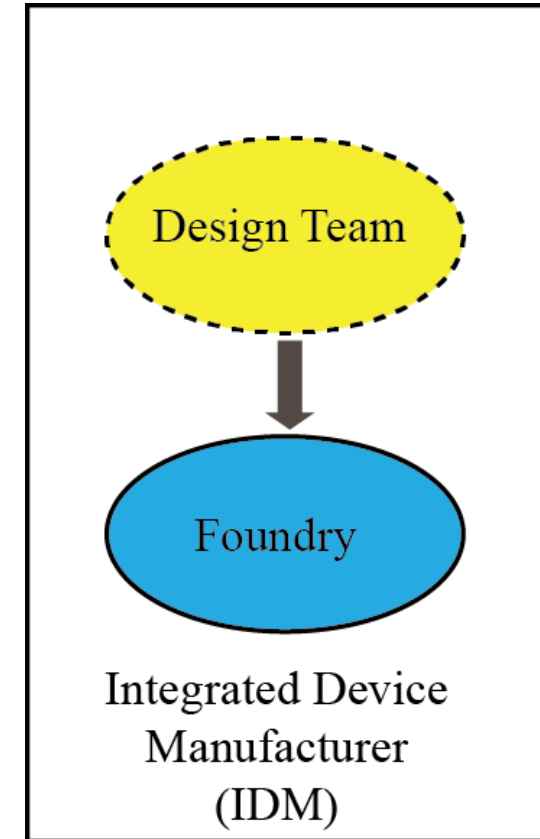
- Only fabrication (for others)
- Draws business from many companies and utilize foundry to full potential .
- Example: TSMC, UMC, GF, etc.



Semiconductor Industry: Business Model (2)

Integrated Device Manufacturers:

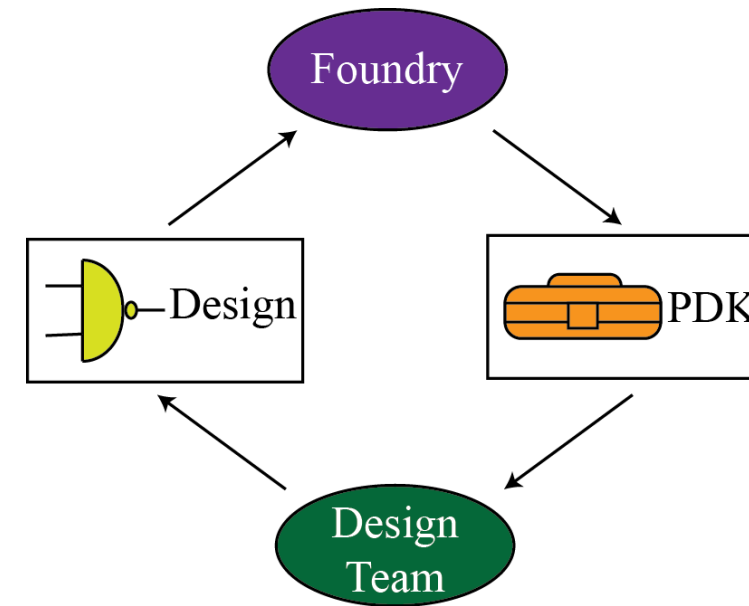
- Both designing and fabrication done in the same company
- Production is more efficient and cost-effective due to control over all the steps of the process.
- Example: Intel, Samsung



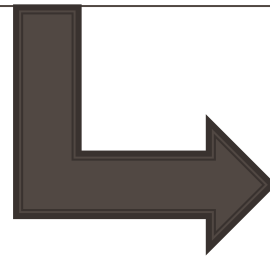
Design and Fabrication

Design and Fabrication:

- Related Task
- Share Information:
 - Process Design Kit (PDK)
 - Design (Layout)



Introduction: Some Basic Concepts



Different VLSI
Design Flows

Different VLSI Design Flow

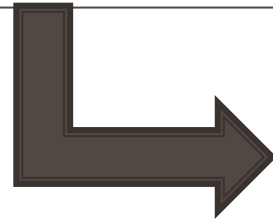
What is VLSI Design Flow?

- Methodology to design an IC such that it “behaves” as per a given functionality

What decides VLSI Design Flow?

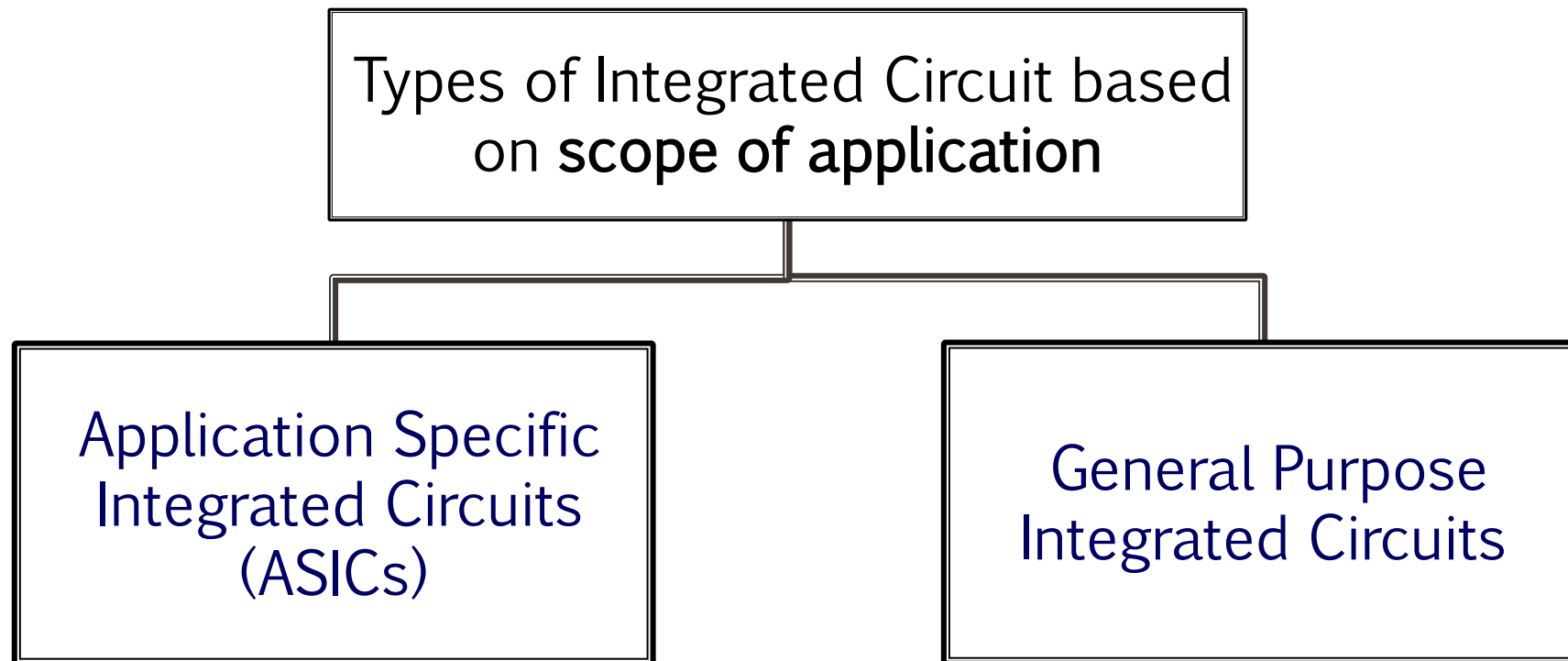
- The VLSI design flows depends on the type of integrated circuits:
 - *Scope* of application
 - *Design Styles*

Introduction: Some Basic Concepts



Types of Integrated
Circuits

Types of Integrated Circuit



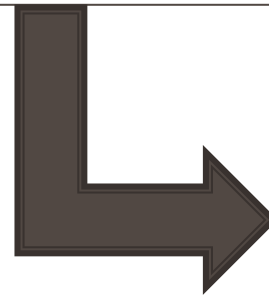
Types of Integrated Circuit

	Application Specific Integrated Circuits (ASICs)	General Purpose Integrated Circuits
Functionality		
Examples		
Programmability		
Volume of production		

Types of Integrated Circuit

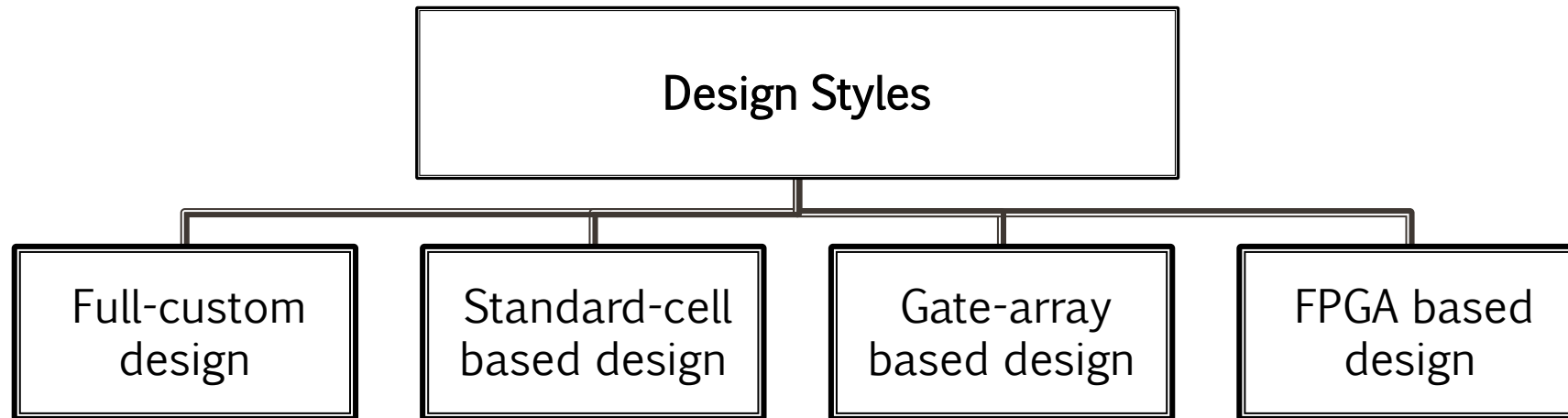
	Application Specific Integrated Circuits (ASICs)	General Purpose Integrated Circuits
Functionality	A chip designed to perform as a particular end system	A chip designed to perform as a wide range of end-system
Examples	IC for digital camera, audio/video processor, security chip etc.	Microprocessors, memory, FPGA
Programmability	Not software programmable to perform a wide variety of different tasks	Usually software programmable to perform a wide variety of different tasks
Volume of production	Less	More

Introduction: Some Basic Concepts



Types of Design
Styles

Types of Design Styles

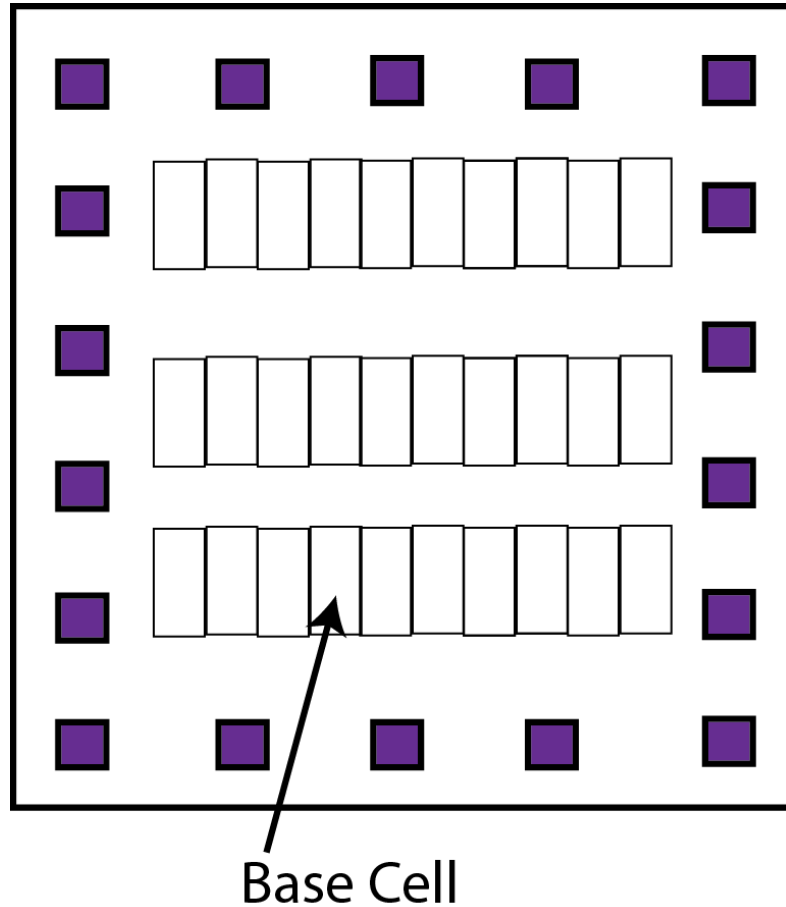


- FPGA: Field Programmable Gate Array

Full-custom design

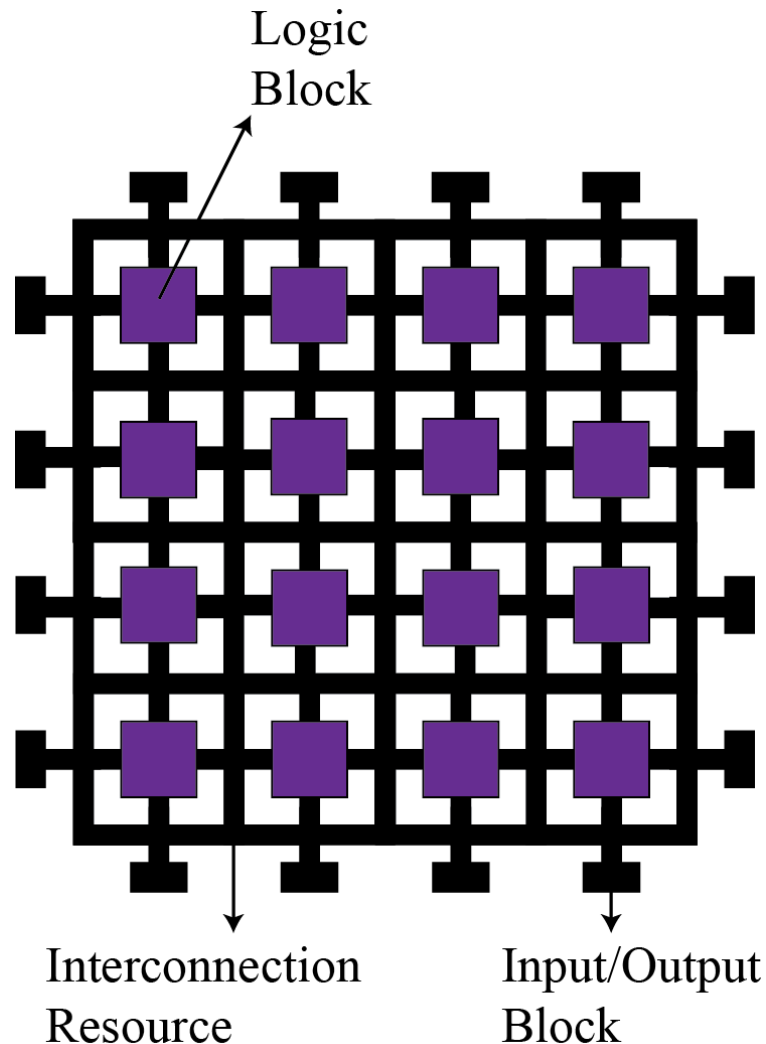
- **Layout** of transistors and interconnects are design specific
- Huge **design effort**
- Very few designs are full-custom
- *Analog mixed/signal* designs
- High **volume** products such as microprocessors (some portion)
- Design can be highly **optimized**

Gate-array based design



- Transistors are *predefined* on the IC in the form of a gate array
- Smallest element that is repeated to form the gate array is called *base cell* or *primitive cell*
- Designer define *only* the interconnection between transistors:
 - Top-most layers of the masks are design specific
- Fixed functionality of the *base cell* may make implementing some functions such as memory difficult or inefficient
- Some custom blocks can be embedded in the IC to obtain special functionality such as memory, micro-controller etc.

FPGA based design

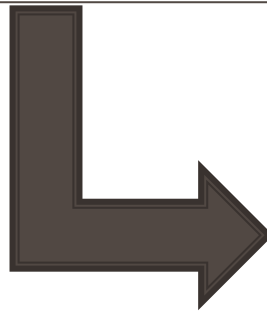


- IC hardware is fixed
 - Designer obtain the desired functionality by programming
 - Programming changes the interconnections between the elements of the circuits
-
- FPGA consist of array of Logic Blocks, IO pads and routing channels
 - Logic Blocks can be programmed to perform different functions such as AND, OR, Adder etc.
 - FPGA may also have embedded microprocessors, analog components and blocks performing special functions such as DSP block
 - Xilinx (AMD), Altera (Intel)

Design Styles (Summary)

	Full-custom design	Standard-Cell based design	Gate-array based design	FPGA based design
Description	Design specific customization at the level of transistors and layout	Pre-characterized cells/macros instantiation and interconnect design specific	Transistors predefined on wafer, interconnect design specific	Programming logic blocks and interconnect
Design effort	Highest	High	Lower	Lower
Custom Mask Layers	All	All	Top few layers	None
Performance, Power, Area	Best	Very Good	Comparatively Inferior	Comparatively Inferior

Introduction: Some Basic Concepts



Economics of
Integrated Circuit

Economics of Integrated Circuits: Components

Fixed Product Cost

- Cost of designing/designer
 - Depends on size/complexity of design
- Software Tools
- Hardware
- Cost of Masks
 - Depends on number of layers

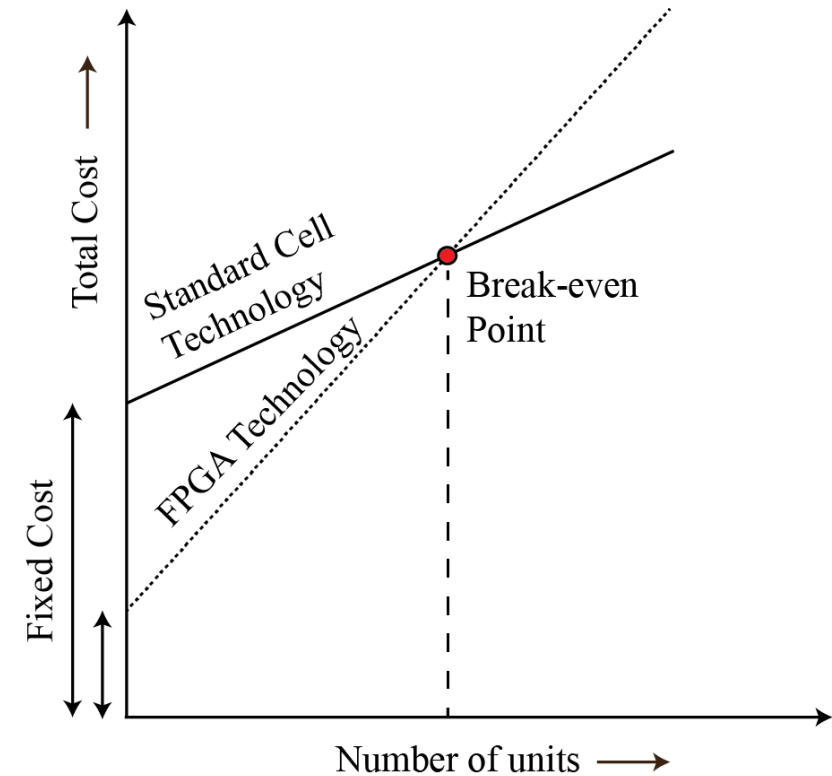
Variable Product Cost

- Cost of wafer
- Cost of die
 - Depends on size of die
 - Yield

$$\begin{aligned} & \text{Total product cost} \\ &= \text{Fixed product cost} + \text{Variable product cost per Product} \times \text{Number of Product} \end{aligned}$$

Economics of Integrated Circuits: Comparison

	Standard-cell Based Design	FPGA Based Design
Fixed Cost	High: Designing Cost, Tools, Mask	Low: Tools for programming
Variable Cost per Product	Low: Cost of Die (Small Die Size, Higher Yield)	High: Cost of Die (Large Die Size, Low Yield)



For small volume FPGA is better, for large volume Standard-cell Based Design is Better.

References

- S. Saurabh, “Introduction to VLSI Design Flow”. Cambridge: *Cambridge University Press*, 2023.